American Computer Science League

2025-26 Finals ● Short Problems ● Junior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$(A + \overline{B})(\overline{B}\,\overline{C}) + A\overline{B}C$$

A. 1
B. 3
C. 5
D. 7
E. None of the above

2. Boolean Algebra

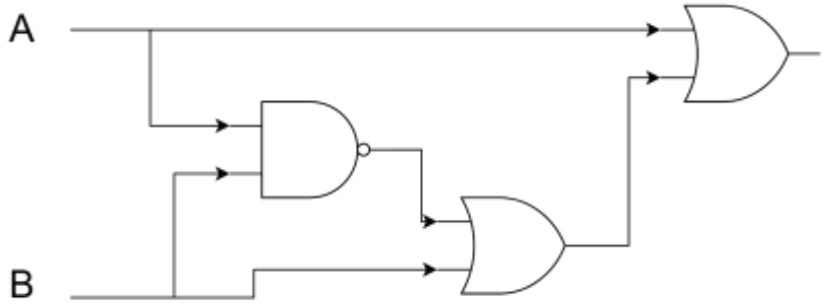
Simplify the following Boolean expression:

$$A(\overline{B} + \overline{AC}) + \overline{A}\,\overline{B}(\overline{C + \overline{A}})$$

A. $\overline{A}B + A\overline{C}$
B. $A\overline{B} + \overline{A}C$
C. $A\overline{B} + A\overline{C}$
D. $\overline{A}B + \overline{A}C$
E. None of the above

3. Digital Electronics

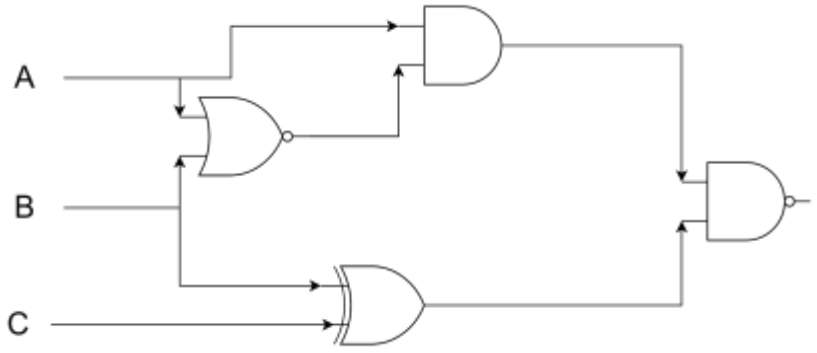
Simplify the Boolean expression for the following digital circuit:



A. 0
B. A
C. B
D. 1
E. None of the above

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 6
D. 8
E. None of the above

5. Bit-String Flicking

Solve for X, a 5-bit string:

(NOT 00011) AND (LCIRC-2 (RSHIFT-2 01011)) OR X = 01101

A. 1*010
B. 0*110
C. 1*001
D. 0*101
E. None of the above

6. Bit-String Flicking

Evaluate the following bit-string expression after converting each octal or hexadecimal number to an 8-bit string in binary:

(RSHIFT-2 $9A_{16}$) OR (RCIRC-2 11000101_2) AND (LCIRC-2 (NOT 361_8))

A. 00110110
B. 00010110
C. 01101110
D. 10011010
E. None of the above

7. Recursive Functions

Find $f(21)$, given:

$$f(x) = \begin{cases} 2 \cdot f(x-1) + 3 & \text{if } x \text{ is a multiple of 3 and 5} \\ f(x-3) + 1 & \text{if } x \text{ is a multiple of 2 or 7} \\ 2x - 4 & \text{otherwise} \end{cases}$$

A. 45
B. 44
C. 43
D. 42
E. None of the above

8. Recursive Functions

Find $f(10, 7)$, given:

$$f(x,y) = \begin{cases} (x-1)(y-1) & \text{if } x = 2y \\ f(y, x-2) - 2 & \text{otherwise} \end{cases}$$

A. 1
B. 3
C. 5
D. 8
E. None of the above

9. Prefix/Infix/Postfix Notation

Translate the following prefix expression to postfix:

+ + a * b − ^ a 2 c * a b

A. aba2^c−*+ab*+
B. aba2^c−*ab*++
C. aba2c^−*+ab*+
D. aba2^c−*+ab+*
E. None of the above

10. Prefix/Infix/Postfix Notation

Define: a & b = max(a,b) and a # b = min(a,b)

Evaluate the following postfix expression if all numbers are single digits:

4 9 8 & 7 3 # + * 2 6 2 # ^ 3 * 5 1 + 4 & / −

A. 24
B. 46
C. 48
D. 50
E. None of the above

11. Computer Number Systems

Convert the following number to base 8. Which digit occurs most often?

$EAFBDC_{16}$

A. 1
B. 3
C. 5
D. 7
E. None of the above

12. Computer Number Systems

Evaluate the following and express the answer in octal.

$2_{16} + 20_8 + 202_{16} + 2026_8$

A. 2250
B. 2742
C. 3052
D. 3072
E. None of the above

13. Data Structures

The Winter Olympics were held in Italy this year.
Build a binary search tree for:

WINTEROLYMPICS

What is its depth?

A. 4
B. 5
C. 6
D. 7
E. None of the above

14. Data Structures

Create a stack by using the following rules for making a Big Mac sandwich using Buns, Sauce, Lettuce, Onions, Cheese, Pickles, and Whole Beef Patties:

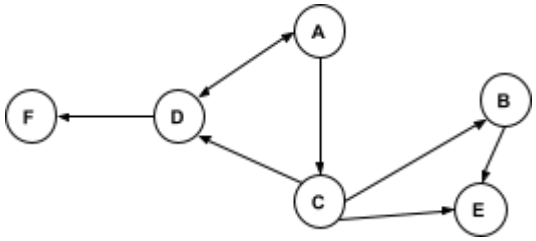
PUSH("B")
PUSH("S")
PUSH("L")
PUSH("O")
PUSH("C")
PUSH("WBP")
PUSH("S")
PUSH("L")
PUSH("P")
PUSH("WBP")
PUSH("B")

How many POP operations followed by PUSH operations would remove the cheese if a person is allergic to dairy?

A. 7 POP and 6 PUSH
B. 6 POP and 7 PUSH
C. 5 POP and 4 PUSH
D. 4 POP and 5 PUSH
E. None of the above

15. Graph Theory

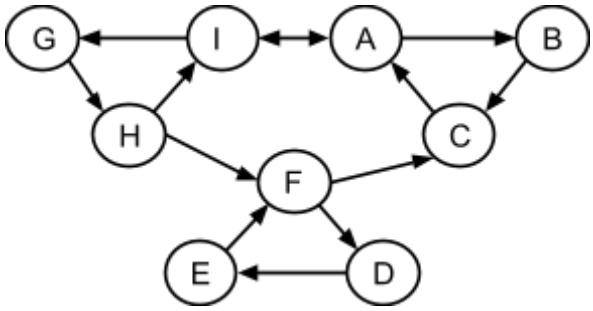
Given the directed graph below, how many paths of length 2 are there that start and end at two different vertices?



A. 6
B. 8
C. 10
D. 12
E. None of the above

16. Graph Theory

How many cycles are in the graph below?



A. 3
B. 4
C. 5
D. 6
E. None of the above

17. What Does This Program Do? (Branching)

In the following program, which of these input values will NOT be output?

```
input num
if num % 2 == 0 then
     if num % 3 == 0 then
          output num
     else
         if num % 4 == 0 then
             output num
         end if
     end if
else
    if num % 5 == 0 then
        output num
    end if
end if
```

A. 18
B. 20
C. 35
D. 70
E. None of the above

18. What Does This Program Do? (Loops)

The value of $\frac{\pi}{4}$ can be approximated by the series

$$\frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \ldots$$

What statement should be inserted into the empty box so that the following program computes the approximate value of π?

```
s = 1
sum = 0
for n = 1 to 10000
     [          ]
     sum = sum + term
     s = -1 * s
next n
pi = sum * 4
output pi
```

A. `term=s/(2*n-1)`
B. `term=s/2*n-1`
C. `term=s*(2*n-1)`
D. `term=s/(2*(n-1))`
E. None of the above

19. What Does This Program Do? (Arrays)

Array **arr** contains 6 integers, indexed 0 through 5, that must be placed into ascending order.

3	5	7	9	1	2

At the end of the 3rd time through the outside loop, what will the value of **arr**(5) be?

```
for x = 0 to 4
     for y = x+1 to 5
          if arr(x) > arr(y) then
               t = arr(x)
               arr(x) = arr(y)
               arr(y) = t
          end if
     next y
next x
```

A. 3
B. 5
C. 7
D. 9
E. None of the above

20. What Does This Program Do? (Strings)

What is the output for the following program?

```
s = "ACSLFINALSCOMPETITION"
ans = ""
for x = 0 to 9
     y = len(s)-x-1
     if s[x] > s[y] then
          ans = ans + s[x]
     else
          ans = ans + s[y]
     end if
next x
output ans
```

A. NOSTITNPMSC
B. NOSTITNPMS
C. ACILFIEALO
D. ACILFIEALOC
E. None of the above